AfyaPlus Generative AI Platform — Engineering Decision Memorandum

To: CTO & Medical Director | From: AfyaPlus Engineering | Date: 11 August 2026 | Re: Week 3 Observability Findings

Executive Summary

Over six weeks in production with zero automated evaluation, drift monitoring, or cost tracking, we have now instrumented all three layers. The headline finding: our two candidate local models (Llama 3.1 8B, Mistral) are clinically **on-topic and judged safe by an LLM reviewer (4.4–5.0/5 average alignment)**, but **0 of 30 evaluated responses cleared our automated quality gate** due to response verbosity, not incorrect content. This blocks the cost-optimised routing strategy modelled in Phase 3 and keeps 100% of traffic on mandatory human-in-the-loop review under current policy.

Quality Performance Breakdown

15 clinical questions across USSD, Mobile App, and Web Portal were tested against both models (30 total evaluations). Both models scored strongly on **LLM-judge alignment (correctness, groundedness, relevance, helpfulness)** at 4.4–5.0/5, but scored 0.02–0.15 on ROUGE-L and Token F1 — well below our 0.70 pass threshold. Root cause: both models return long, numbered, safety-caveated answers (e.g. “Consult a healthcare professional” disclaimers, multi-step formatting) that diverge lexically from our terse reference answers, even when clinically correct. This is a **metric-design limitation, not a model-capability failure** — but it means our current automated gate cannot yet be trusted to pass safe answers on its own.

Cost & Efficiency Analysis

Simulated 30-day spend across the three channels totals **$75.78** (30,000 requests, 75% Llama 3.1 / 25% Mistral canary split). Separately, trimming the system prompt from a 200-token chain-of-thought instruction to a 30-token direct instruction cuts per-request cost by **12.1%** on both models — a low-risk, immediate saving that doesn't depend on the quality-gate issue above. However, the **structural routing savings we modelled (auto-routing each feature to its cheapest quality-passing model) currently total $0**, precisely because no model passes the gate for any feature yet. This reframes the $0 figure: it is the quantified cost of mandatory human review, not a failed optimisation.

30-Day Simulated Spend (30,000 req, 75/25 model split)	
USSD	$26.41
Web Portal	$24.94
Mobile App	$24.43
Total	$75.78

Systemic Operational Risks

Drift: Simulated 3-month production data shows statistically significant drift (two-sample KS-test, $p<0.001$) first appearing in **Month 2 on ROUGE-L**, with response-quality proxy declining 0.80→0.75→0.70 and latency rising 954ms→1451ms over the same window. Automated alerting correctly flagged the exact month and column. **Quality gate:** as above, the gate is currently mis-calibrated for verbose-but-correct clinical responses, creating risk of rejecting safe answers on formatting grounds alone if this gate were used to auto-approve in the future. **Environment:** local development required disabling endpoint security (Kaspersky) to permit file writes — documented in README, low risk but worth IT awareness before this pipeline runs unattended in CI.

Actionable Engineering Roadmap

- **1. Re-calibrate the quality gate** to weight LLM-judge alignment above ROUGE-L/F1 for conversational clinical responses, or constrain the system prompt toward concise, reference-matching output. *Justification:* unlocks the $0-blocked structural cost routing in Phase 3 and reduces unnecessary 100% human-review load — direct cost and clinician-time savings once resolved.
- **2. Adopt the trimmed direct-prompt system instruction immediately** (12.1% per-request savings, no dependency on the gate fix). *Justification:* lowest-risk, fastest-to-ship saving identified this cycle.
- **3. Establish a monthly drift-review cadence** using the automated alert log (already flags exact month + column), with a defined response SLA once real production drift crosses the same statistical threshold demonstrated here. *Justification:* catches quality degradation before it reaches patients, at zero marginal engineering cost beyond the pipeline already built.